

FruitEnsemble: MLLM-Guided Arbitration for Heterogeneous ensemble in Fine-Grained Fruit Recognition

Enhui Yu¹, Junhui Li^{1,2}, Ruitong Lu¹, Jialu Li³, Youshan Zhang^{2,*}

¹University of Science and Technology Liaoning, ²Chuzhou University, ³Yeshiva University

*Corresponding author, youshan_zhang@chzu.edu.cn

Abstract

Fine-grained fruit classification is a critical yet challenging task in agricultural computer vision, primarily hindered by a severe shortage of high-quality datasets and the high visual similarity between classes. To address these challenges, we first constructed a comprehensive dataset comprising 306 fruit categories with 116,233 samples. Moreover, we propose FruitEnsemble, a practical two-stage dynamic inference framework designed to overcome the generalization limitations of static single-model architectures. In the first stage, FruitEnsemble employs a validation-calibrated weighted ensemble of heterogeneous backbones to generate a robust Top-3 candidate pool. To tackle difficult samples, we introduce an expert arbitration mechanism: when ensemble confidence falls below 0.6, a multimodal large language model (MLLM) is triggered to perform rigorous visual verification by integrating external botanical descriptions using Chain-of-Thought (CoT) reasoning. Furthermore, we optimized the training pipeline with a hard sample-aware joint loss. Extensive experiments demonstrate that FruitEnsemble achieves a classification accuracy of 70.49% and outperforms existing state-of-the-art models. Our framework provides an efficient, deployment-oriented solution for real-world agricultural visual sorting and quality inspection tasks.¹

1. Introduction

Accurate fruit cultivar recognition plays a crucial role in modern agricultural supply chains, directly influencing quality assurance, pricing, and post-harvest storage [13]. Beyond these core tasks, reliable identification also supports downstream applications like automated sorting, yield estimation, and inventory management.

In practice, fruit classification has long relied on manual inspection and expert judgment. While workable at a small

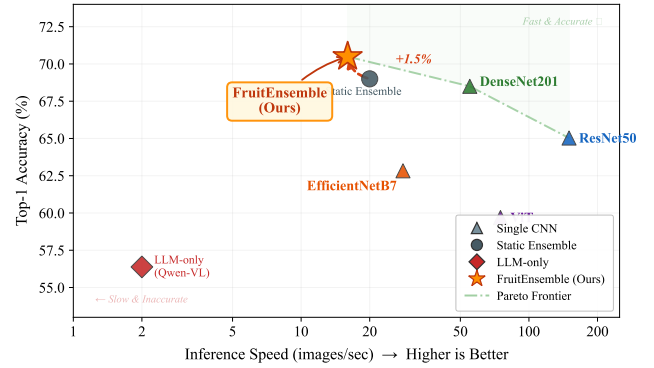


Figure 1. **Accuracy vs. inference efficiency** on the Fruit-306 test set. Single CNNs ($\triangle$) are efficient but limited in accuracy. The LLM-only baseline ($\diamond$) is slow and performs poorly without domain-specific training. Static ensembles ($\bullet$) improve accuracy but remain sub-optimal. FruitEnsemble ($\star$) achieves the best trade-off by invoking the LLM only for uncertain samples.

scale, manual sorting is time-consuming, labor-intensive, costly, and often inconsistent, making it difficult to meet the increasing demand for automation in large-scale agricultural systems. Image-based fruit classification has emerged as an attractive alternative because it is non-destructive, and easily integrated into intelligent sorting lines, harvesting robots, and agricultural monitoring platforms.

Despite recent advances in modern visual backbones, fruit cultivar recognition remains a challenging fine-grained visual classification problem [25]. Cultivars such as Red Fuji and Gala may appear very similar in color, shape, and size, yet they must be reliably distinguished in applications such as automated sorting and quality control. The challenge arises from both subtle inter-class differences and significant intra-class variation caused by factors such as ripeness, illumination, viewpoint, and occlusion. Additionally, agricultural datasets are often long-tailed, with a few common cultivars dominating the training data while many rare ones are underrepresented [10]. Existing visual methods are efficient and relatively mature, but a single model is

¹This research was funded by the Research Projects of Anhui Province Education Department (Grant No. 2025AHGXZK20009).

often not robust enough across all fine-grained categories. Ensemble methods can improve robustness [8], yet they still process every sample in the same way and cannot adapt their computation to sample difficulty.

Vision-language models have recently shown strong capabilities in image understanding and cross-modal reasoning. However, in fruit classification, prior work has focused primarily on pure visual recognition based on CNNs [22], Vision Transformers [9], or their ensembles. The use of language-guided reasoning with category-level descriptions for fine-grained fruit recognition remains relatively under-explored. As a result, neither pure visual classification nor indiscriminate use of large models is ideal: the former can struggle with fine-grained and long-tailed cases, while the latter is expensive to deploy and requires sufficient domain grounding for reliable predictions [42]. As shown in Fig. 1, existing paradigms still face a clear trade-off between recognition accuracy and inference efficiency, which motivates our approach.

To address these challenges, we propose **FruitEnsemble**, a dynamic two-stage framework for fine-grained fruit classification. In the first stage, four complementary visual models – ResNet50 [17], DenseNet201 [21], EfficientNetB7 [34], and Vision Transformer – are combined through confidence-aware weighted fusion to produce a prediction and a confidence score. In the second stage, a multimodal LLM (Qwen-VL-Plus) is invoked only when the ensemble is uncertain and performs reasoning over the top candidate classes with the aid of curated category descriptions [46]. This design allows the easiest samples to be handled efficiently by the visual ensemble, while reserving more expensive language-guided reasoning for genuinely ambiguous cases. Our contributions are three-fold:

- We construct **Fruit-306**, a large-scale fine-grained fruit recognition benchmark with 306 categories and 116,233 images. Uniquely, it pairs visual data with expert-curated *textual morphological descriptions*, enabling rigorous evaluation of knowledge-guided multimodal reasoning in agriculture.
- We present **FruitEnsemble**, a practical two-stage recognition framework that combines validation-calibrated heterogeneous ensembling with selective MLLM-based arbitration. By constraining the MLLM to a *Top-K* candidate set and providing category-specific botanical descriptions, the system improves hard-case verification while avoiding unrestricted open-set generation.
- We propose a *Hard Sample-Aware Joint Optimization* strategy. By applying a specific **diversity loss** only to hard samples that the router has detected, we force backbones to learn complementary features where needed, while also optimizing the training process. This enables SOTA **70.49% accuracy** with real-time latency (**19.8 ms**), invoking the LLM for only 15% of cases.

2. Related Work

2.1. Fine-Grained Fruit Recognition

Fine-grained visual recognition has been studied extensively in domains such as birds and cars. Early approaches emphasized part discovery and alignment [45], later methods relied on attention mechanisms [11], and recent transformer-based architectures have shown strong performance [16]. However, transferring these methods to fruit variety recognition poses unique challenges: cultivar-level differences are rarely expressed through stable semantic parts but instead through weak cues such as texture, gloss, stripe density, and color transition [20], all of which are easily affected by lighting, occlusion, and growth stage. Deep learning has been applied to apple cultivar classification [4] and citrus grading [40], yet most work addresses coarse-grained separation among a small number of varieties and lacks sufficient discriminative power when facing hundreds of visually similar cultivars.

Multimodal large language models such as Qwen-VL-Plus and GPT-4V [41] provide a unified interface for visual understanding and language-based reasoning, and can generate interpretable decisions through chain-of-thought prompting [38]. Initial studies have evaluated the zero-shot capability of MLLMs for crop disease diagnosis [37], but their performance on cultivar-level fine-grained discrimination remains unstable, and inference latency makes them impractical as standalone high-throughput classifiers.

2.2. Agricultural Vision Benchmarks

The benchmark landscape further compounds this limitation. PlantVillage [27] focuses on disease recognition, and Fruits-360 [28] provides images under controlled backgrounds for fruit-level classification; neither captures the difficulty of variety-level recognition under realistic conditions. In particular, existing public benchmarks rarely combine a large number of similar cultivar categories, substantial intra-class variation — the latter being essential for vision-language joint reasoning and a gap that our dataset is specifically designed to address.

2.3. Ensemble Model

Ensemble learning improves robustness by combining predictions from multiple models [23] and has been applied to crop disease recognition [1] and fruit quality grading. However, conventional ensembles are static: every sample is processed by every model with fixed fusion rules. In high-throughput scenarios such as automated fruit sorting lines, this uniform computation is inefficient since most samples are easy to classify. Adaptive inference methods such as early-exit networks [35] and dynamic ensemble selection [7] adjust computation per sample, but they are mostly limited to homogeneous model families and have

not been systematically applied to agricultural fine-grained recognition, where lightweight classifiers and stronger reasoning models need to collaborate.

Our work targets closed-set recognition over hundreds of visually similar fruit varieties and combines three elements: a heterogeneous ensemble of complementary vision backbones, a confidence-based dynamic escalation mechanism, and category-level textual descriptions for semantic grounding during MLLM reasoning. The MLLM serves as a selective arbiter for ambiguous samples rather than a replacement for specialized visual classifiers.

3. Fruit-306 Dataset

We create **Fruit-306**, a benchmark for fine-grained fruit recognition in realistic settings. The dataset contains **116,233** images from **306** categories and is designed to support both standard visual classification and vision-language reasoning. Compared with existing fruit datasets, Fruit-306 is characterized by three aspects that are central to this work: a larger set of fine-grained categories, real-world visual complexity, and category-level textual descriptions. Images are collected under diverse conditions and include cluttered backgrounds, viewpoint changes, occlusion, and different ripeness stages, making the benchmark substantially more challenging than other datasets.

Fruit-306 adopts a fixed **7:1:2** split into training, validation, and test sets, containing **81,233**, **11,488**, and **23,522** images, respectively, while preserving the original class distribution. As shown in Table 1, the dataset exhibits a clear long-tail distribution: the largest category contains **1,276** images (*jaboticaba*), while the smallest contains only **25** images (*muscadine-grape*), resulting in an imbalance ratio of **50:1**. This naturally preserved long-tail distribution reflects real-world agricultural data availability and requires reliable recognition for both frequent and rare categories.

Table 1. Statistical breakdown of the **Fruit-306** dataset. The partition maintains the original long-tail distribution across training, validation, and testing sets. Note the significant variance in class sizes, ranging from abundant commercial varieties to rare species.

Split	Ratio	# Images	# Classes	Max Sample Class	Min Sample Class	Avg. Size
Training	70%	81,223	306	1,276	25	379.85
Validation	10%	11,488		(jaboticaba)	(muscadine_grape)	
Testing	20%	23,522				
Total	100%	116,233	306	<i>Imbalance Ratio (Max/Min): 50:1</i>		

Beyond image labels, each category is paired with a short expert-curated textual description that summarizes its discriminative visual traits, including morphology, color patterns, texture, and local structure. These descriptions provide category-level priors that are useful when visual evidence alone is insufficient. In our framework, they serve as semantic guidance for the language model when the visual ensemble is uncertain, directly supporting the dynamic

inference strategy introduced in Sec. 4.

The class distribution is summarized in Fig. 2. The histogram and cumulative curve confirm a severe long-tail pattern, with a small number of head classes accounting for a large fraction of the data. This imbalance, together with real-world appearance variation, makes Fruit-306 a more demanding benchmark than balanced fruit datasets.

The dataset follows standard PyTorch `ImageFolder` format, with folders for training, validation, and test splits. This structure is used in all experiments.

Table 2 compares Fruit-306 with representative fruit and VegFru [19] datasets. Fruit-306 combines fine-grained category coverage, realistic image conditions, natural long-tail imbalance, and textual descriptions in a single benchmark. This combination is particularly suitable for evaluating the proposed method, which integrates visual experts with language-guided reasoning.

Table 2. **Comparison of Fruit-306 with existing fruit related datasets.** Our dataset leads in classes, realism, and multimodal annotations (Desc.: Description).

Dataset	# Classes	# Images	Real-World	Long-Tail	Text Desc.
Fruits-360 [14]	131	90k	No	No	No
VegFru [19]	292	160k+	Yes	Yes	No
FruitVision [5]	5	10,154	Yes	No	No
Fruit-306 (Ours)	306	116,233	Yes	Yes	Yes

From the recognition perspective, Fruit-306 is challenging for four reasons: high inter-class similarity, large intra-class variation, cluttered natural backgrounds, and severe class imbalance. Figure 3 shows typical examples. These factors motivate the design of **FruitEnsemble**: easy samples are efficiently classified by the ensemble vision model, while ambiguous cases benefit from language-guided reasoning with category descriptions. We evaluate this strategy on the fixed split of Fruit-306 in Sec. 5.

4. Method

Fine-grained fruit variety recognition faces two intertwined challenges: visually similar cultivars demand strong discriminative power, while long-tail class distributions and high-throughput deployment requirements impose strict efficiency constraints. To address this trade-off, we present **FruitEnsemble**, a hierarchical collaborative framework that uses complementary inductive biases of heterogeneous visual architectures for robust feature learning. We only apply computationally heavy LLM reasoning to ambiguous examples that cannot be resolved by visual information alone. Formally, let $\mathcal{X}$ denote the input image space and $\mathcal{Y} = \{1, \dots, C\}$ denote the set of $C = 306$ fruit categories. Our framework $\mathcal{F}$ is defined as a tuple $(\mathcal{M}, \mathcal{A}, \mathcal{R}, \Phi_{\text{LLM}})$, comprising: (1) a heterogeneous backbone ensemble $\mathcal{M}$; (2) an uncertainty-aware aggregation operator $\mathcal{A}$; (3) a

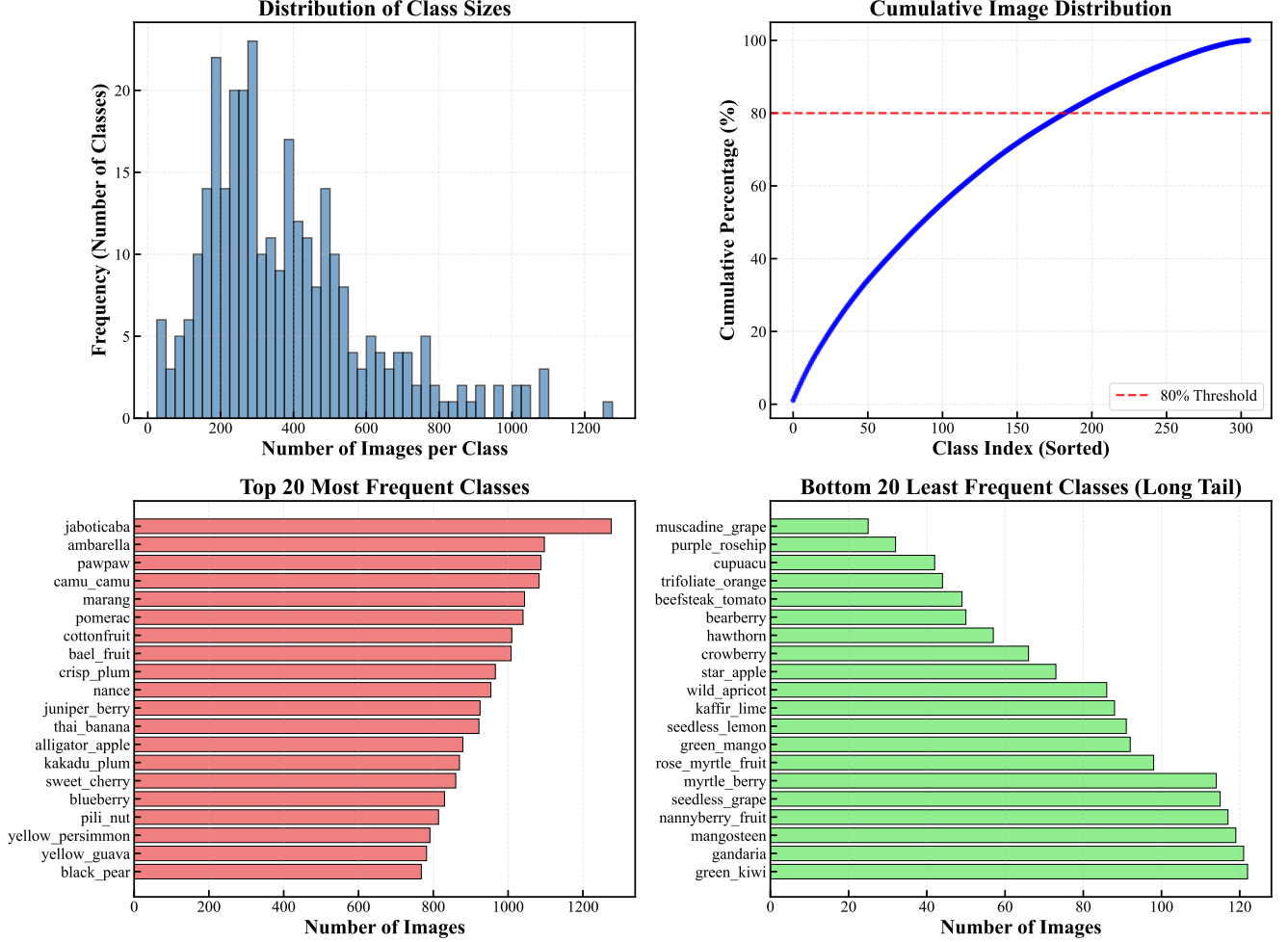


Figure 2. **Comprehensive Statistical Analysis of Fruit-306.** (a) Histogram of class sizes showing the frequency distribution of samples per category. (b) Cumulative distribution curve indicating that the top 20% of classes account for a significant majority of the total images. (c) Top 20 most frequent classes (Head), dominated by common commercial varieties. (d) Bottom 20 least frequent classes (Tail), highlighting the data scarcity in rare species. These statistics confirm the severe long-tail nature of the dataset.

confidence-gap triggered router $\mathcal{R}$; (4) a Top- K constrained LLM arbiter Φ_{LLM} . The overall pipeline is illustrated in **Figure 4**.

4.1. Uncertainty-Aware Heterogeneous Fusion

To mitigate the inductive bias inherent in single architectures, we construct an ensemble $\mathcal{M} = \{m_1, m_2, m_3, m_4\}$ consisting of four distinct backbones [8], selected based on their empirical complementarity (see Table 3):

- **DenseNet201** [21] (m_1): excels in capturing fine-grained texture patterns (e.g., skin speckles) due to dense connectivity.
- **EfficientNetB7** [34] (m_2): provides superior multi-scale representation, yielding high Top-5 recall.
- **Vision Transformer** [9] (m_3): models long-range dependencies and global morphology, offering an orthogonal perspective to CNNs.

- **ResNet50** [17] (m_4): serves as a robust baseline with balanced hierarchical feature extraction.

Architecture-Specific Adaptation. To ensure optimal convergence of these heterogeneous components, we employ tailored training strategies for each backbone, including *topology-aware layer freezing* [17], *gradient clipping with non-finite handling* [15], and *activation-matched classifier heads*. These techniques stabilize training and maximize the individual capacity of each model, forming a robust foundation for the ensemble. Detailed implementation protocols are provided in Sec. 5.

Given an input image x , each backbone m_i outputs a class probability distribution $P_i(x)$. Instead of static averaging, we propose an *Uncertainty-Aware Dynamic Aggregation* mechanism. Let $\mathcal{U}_i(x) = -\sum_c P_i(x)_c \log P_i(x)_c$ denote the predictive entropy [12]. The dynamic weight

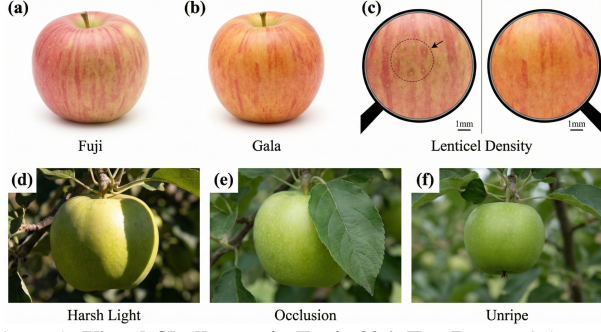


Figure 3. **Visual Challenges in Fruit-306.** **Top Row:** High *inter-class* similarity. Examples of visually similar pairs (e.g., *Fuji* vs. *Gala*) where discrimination relies on subtle textual attributes (e.g., lenticel density). **Bottom Row:** High *intra-class* variance. The same variety shown under different lighting, occlusion, and ripeness conditions. These complexities necessitate dynamic reasoning beyond static classification.

$\alpha_i(x)$ is computed as:

$$\alpha_i(x) = \frac{\exp(-\mathcal{U}_i(x)/\tau_{temp})}{\sum_{j=1}^4 \exp(-\mathcal{U}_j(x)/\tau_{temp})}, \quad (1)$$

where τ_{temp} is a temperature parameter [18]. The aggregated system distribution is $P_{sys}(x) = \sum_{i=1}^4 \alpha_i(x)P_i(x)$. From $P_{sys}(x)$, we extract the Top- K ($K = 3$) candidate set $\mathcal{C}_{top}(x) = \{c_1, c_2, c_3\}$, where c_1 is the most probable class.

4.2. Dynamic Confidence-Gap Arbitration Routing

A distinguishing feature of FruitEnsemble is the *Dynamic Confidence-Gap Arbitration*. Relying solely on the maximum probability $S(x) = P_{sys}(x)_{c_1}$ can be misleading: a model might be confidently wrong, or it might be highly uncertain between two visually similar classes (e.g., two apple varieties) despite having a high P_{c_1} . To capture this ambiguity, we define the **Confidence Gap** [3] $\Delta(x)$ as the margin between the top two predictions:

$$S(x) = P_{sys}(x)_{c_1}, \quad \Delta(x) = P_{sys}(x)_{c_1} - P_{sys}(x)_{c_2}. \quad (2)$$

The routing function $\mathcal{R}(x)$ triggers the LLM arbitration path if the sample is either *uncertain* (low $S(x)$) or *ambiguous* (small $\Delta(x)$):

$$\mathcal{R}(x) = \mathbb{I}(S(x) < \tau_{conf} \vee \Delta(x) < \tau_{gap}), \quad (3)$$

where τ_{conf} and τ_{gap} are empirically determined thresholds. This dual-criterion ensures both low-confidence and *highly contested* samples are escalated for deeper reasoning, preventing the ensemble from making hasty decisions on fine-grained boundaries [11].

The final prediction $\hat{y}$ is determined by:

$$\hat{y} = \begin{cases} c_1, & \text{if } \mathcal{R}(x) = 0 \quad (\text{Direct Path}) \\ \Phi_{LLM}(x, \mathcal{C}_{top}(x)), & \text{if } \mathcal{R}(x) = 1 \quad (\text{Arbitration Path}). \end{cases} \quad (4)$$

Empirically, this strategy routes approximately 15% of the most challenging samples to the LLM, optimizing the cost-accuracy trade-off.

4.3. Top- K Constrained Chain-of-Thought Arbitration

For samples routed to the arbitration path, the LLM acts not as a generative classifier over the entire label space $\mathcal{Y}$, but as a verifier within the restricted hypothesis space $\mathcal{C}_{top}(x)$. This constraint reduces hallucination risks by limiting the solution space to the most probable candidates identified by the visual ensemble. Specifically, let $\mathcal{C}_{top}(x) = \{c_1, \dots, c_K\}$ denote the top- K candidate set output by the preceding stage, where $K \ll |\mathcal{Y}|$. We construct the LLM prompt using the input image, the candidate set $\mathcal{C}_{top}(x)$, and botanical priors $\mathcal{T} = \{T_c\}_{c \in \mathcal{C}_{top}}$, where each T_c is an expert-curated textual description of a candidate fruit category. The LLM Φ_{LLM} performs candidate-aware reasoning:

- Visual Attribute Extraction:** Identify discriminative features $A = \text{Extract}(V(x))$ (e.g., skin glossiness, calyx structure) [29].
- Prior Matching & Exclusion:** Compute semantic alignment scores $M_c = \text{Similarity}(A, T_c)$ and explicitly penalize contradictions:

$$\hat{y}_{llm} = \arg \max_{c \in \mathcal{C}_{top}} (M_c - \lambda_{pen} \cdot \text{Conflict}(A, T_c)). \quad (5)$$

Implementation Details: To operationalize this reasoning process, our system dynamically retrieves the corresponding expert botanical priors from our curated text database. Rather than relying on a static prompt, these retrieved descriptions are injected into a standardized Chain-of-Thought (CoT) template. The MLLM is explicitly instructed to execute the aforementioned extraction and matching steps, finally outputting the decision in a strict JSON format to prevent parsing errors and hallucinations. For the complete prompt templates, extensive database snippets, and full reasoning case studies, please refer to our project repository: <https://mybkgjvgnnd.github.io/Fruit-306-Dataset/>.

4.4. Hard Sample-Aware Joint Optimization

Training such a heterogeneous system requires balancing individual accuracy with collective diversity, particularly for long-tail classes. We propose a *Hard Sample-Aware Joint Loss* [24] $\mathcal{L}_{total}$:

$$\mathcal{L}_{total} = \underbrace{\sum_{i=1}^4 \mathcal{L}_{focal}(P_i, y)}_{\mathcal{L}_{ind}} + \lambda_1 \underbrace{\mathcal{L}_{focal}(P_{sys}, y)}_{\mathcal{L}_{global}} + \lambda_2 \underbrace{\mathcal{L}_{div}^{hard}}_{\mathcal{L}_{div}}, \quad (6)$$

where $\mathcal{L}_{focal}$ addresses class imbalance.

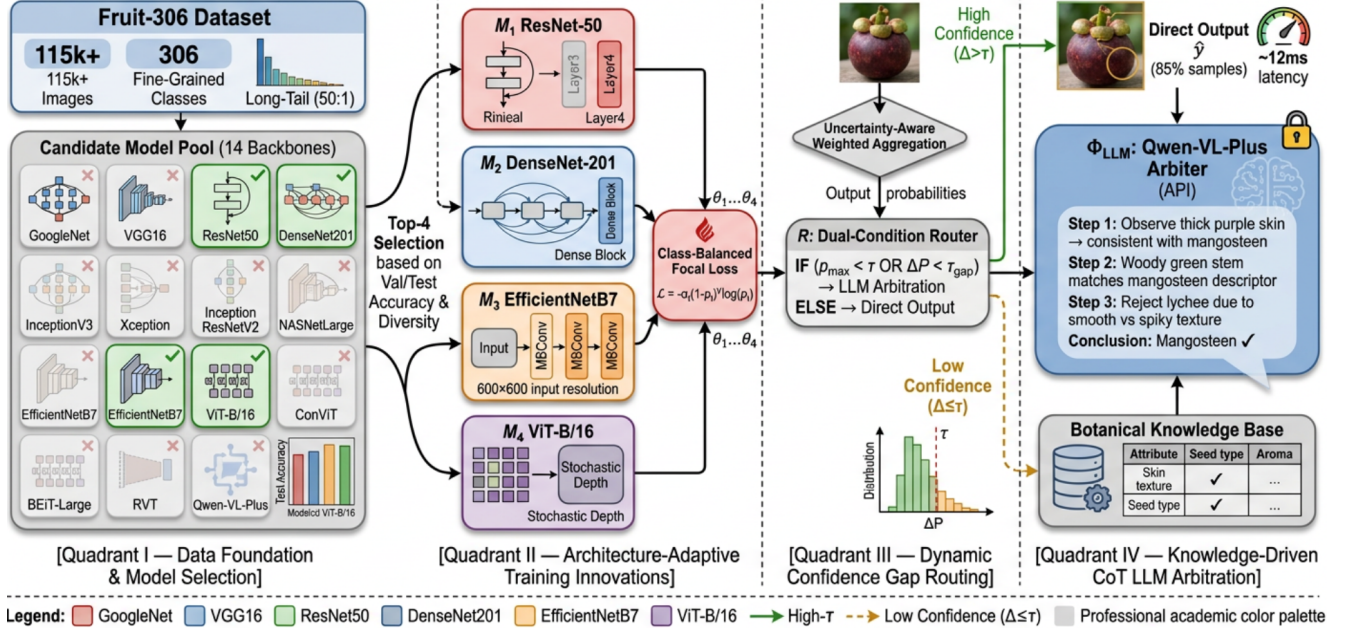


Figure 4. **Overview of the FruitEnsemble Framework.** Input fruit images are first processed by a curated set of four heterogeneous backbone models (M_1 - M_4 : ResNet50, DenseNet201, EfficientNetB7, ViT-B/16) with architecture-adaptive training strategies. Their output probabilities are fused via an *Uncertainty-Aware Weighted Aggregation* module ($\mathcal{A}$) to generate top- k candidate predictions and a confidence gap metric ($\Delta = p_1 - p_2$). The *Confidence-Gap Router* ($\mathcal{R}$) applies a dual-condition rule: if the maximum confidence ($p_{\max} > \tau$) and confidence gap ($\Delta > \tau_{\text{gap}}$), samples follow the direct path for immediate prediction ($\hat{y}$); ambiguous samples (low confidence or small margin) are routed to the *Qwen-VL-Plus LLM Arbitrator* (Φ_{LLM}). The LLM performs Knowledge-Augmented Chain-of-Thought (CoT) reasoning using curated fruit descriptor knowledge to produce the final error-corrected prediction ($\hat{y}^*$).

The critical component is the diversity term $\mathcal{L}_{\text{div}}^{\text{hard}}$, which applies Jensen-Shannon divergence maximization *only* to samples identified as hard by the router ($\mathcal{R}(x) = 1$):

$$\mathcal{L}_{\text{div}}^{\text{hard}} = \frac{1}{|\mathcal{B}_{\text{hard}}|} \sum_{x \in \mathcal{B}_{\text{hard}}} \left[- \sum_{i \neq j} \text{JS}(P_i(x) \parallel P_j(x)) \right], \quad (7)$$

where $\mathcal{B}_{\text{hard}} = \{x \in \mathcal{B} \mid \mathcal{R}(x) = 1\}$. **Rationale:** Applying diversity loss to easy samples may destabilize the consensus on clear patterns. By restricting it to hard samples, we specifically force the ensemble to explore complementary feature subspaces (e.g., texture vs. shape) for ambiguous cases. This ensures that the ground truth label is highly likely to be included in the Top- K candidate set $\mathcal{C}_{\text{top}}(x)$, thereby maximizing the success rate of the subsequent LLM arbitration.

4.5. Robust Training Algorithm

We propose a *Robust High-Resolution Fine-Tuning* strategy (Algorithm 1) that integrates heterogeneous adaptation with fault-tolerant optimization. Key features include: (1) **Topology-Aware Freezing** [43], which unfreezes only final decision blocks to preserve generic features; (2) **Fault-Tolerant Flow**, employing NaN-aware batch skipping and

non-finite gradient clipping to prevent collapse on noisy data; (3) **EMA Regularization**, maintaining weight averages as a temporal ensemble; [44] (4) **Dynamic Scheduling** via Cosine Annealing with Warm Restarts to escape local minima.

4.6. Computational Efficiency

Concerns regarding LLM-augmented ensemble latency are mitigated by our *adaptive computation*. Let T_{CNN} and T_{LLM} denote inference times for backbones and the LLM, respectively. Unlike naive ensembles ($4T_{\text{CNN}} + T_{\text{LLM}}$), our Dynamic Confidence-Gating invokes the LLM only for a fraction $\gamma \approx 0.15$ of samples. The expected latency is $E[T] = 4T_{\text{CNN}} + \gamma T_{\text{LLM}}$. Since backbones run in parallel and $\gamma \ll 1$, FruitEnsemble achieves near-ensemble accuracy with latency comparable to a single model, ensuring viability for real-time deployment.

5. Experiments

5.1. Experimental Setup

All experiments were conducted on a workstation equipped with an NVIDIA GeForce RTX 5060 Ti GPU using PyTorch with CUDA acceleration. To construct a heterogeneous ensemble with complementary representa-

Algorithm 1 Robust High-Resolution Fine-Tuning

Require: Dataset $\mathcal{D}$, Backbones $\{\Theta_{base}^{(i)}\}$, Resolutions $\{R_i\}$
Ensure: Optimized Ensemble $\{\Theta^{*(i)}\}$, EMA Shadows $\{\Theta_{ema}^{(i)}\}$

```
1: Phase 1: Adaptation
2: for each backbone  $i$  do do
3:   Resize inputs to  $R_i$ ; Unfreeze final block & classifier
4:   Init  $\Theta^{(i)} \leftarrow \Theta_{base}^{(i)}$ ,  $\Theta_{ema}^{(i)} \leftarrow \Theta^{(i)}$ 
5: end for
6: Phase 2: Optimization ( $\mathcal{L}_{total}$ , AdamW, Cosine Scheduler)
7: while epoch  $e < E_{max}$  do do
8:   for batch  $(X, y)$  do do
9:     Compute outputs  $\{P_i\}$ ,  $P_{sys}$ ; Identify hard samples  $\mathcal{B}_{hard}$ 
10:     $\mathcal{L} \leftarrow \mathcal{L}_{total}(\{P_i\}, P_{sys}, y, \mathcal{B}_{hard})$ 
11:    if  $\mathcal{L}$  is NaN/Inf then
12:      Skip batch; continue
13:    end if
14:     $\mathcal{L}.backward()$  (with accumulation)
15:    if update step then then
16:      Clip gradients (tolerant); Optimizer step(); Zero grad
17:      for each  $i$  do do
18:        Update EMA:  $\Theta_{ema}^{(i)} \leftarrow \lambda \Theta_{ema}^{(i)} + (1 - \lambda) \Theta^{(i)}$ 
19:      end for
20:    end if
21:  end for
22: end while
23: return Best checkpoints
```

tion capabilities, four widely used backbones were selected: ResNet50, DenseNet201, EfficientNetB7, and Vision Transformer (ViT-B/16). These architectures represent diverse design paradigms in fine-grained visual classification (FGVC): ResNet50 provides robust hierarchical feature extraction, and DenseNet201 captures fine-grained texture patterns through dense connectivity. EfficientNetB7 offers a strong multi-scale representation with compound scaling, and Vision Transformer models long-range dependencies via global self-attention.

Training was performed for 100 epochs using the AdamW optimizer with an initial learning rate of 5×10^{-5} and weight decay of 0.01. The batch size was set to 8. To address class imbalance in the dataset, Focal Loss ($\gamma = 2.0$) was employed, where the class balancing parameter α was set inversely proportional to class sample frequency. Input images were resized to 224×224 for most models, while adhering to the original architecture specifications, 600×600 for EfficientNetB7, followed by normalization to the range $[0, 1]$. Standard data augmentation techniques were applied, including random horizontal flipping, random cropping, color jittering, and $\pm 15^\circ$ rotation.

Different fine-tuning strategies were applied to each backbone to better adapt them to the target task. For ResNet50, the final residual block (layer4) and classification head were unfrozen for training. DenseNet201 unfroze the final dense block and applied an exponential moving average (EMA) to stabilize training. EfficientNetB7 fine-tuned the last MBConv block with cosine annealing

scheduling, while Vision Transformer employed stochastic depth regularization to improve generalization.

Model performance was evaluated using Top-1 accuracy, Top-5 accuracy, and macro-F1 score on validation and test sets. Additionally, performance across three fruit quality grades (high-quality, defective, and flawed) was analyzed. For the ensemble framework, the aggregation temperature was set to 1.0, the routing confidence threshold was 0.60, and the candidate set size was $K = 3$. The LLM arbitrator (Qwen-VL-Plus) was prompted using chain-of-thought (CoT) reasoning, with maximum generation length 512 and temperature 0.7. A response caching mechanism was used to reduce repeated inference latency.

To facilitate reproducibility and future research, the source code and the proposed Fruit-306 dataset are available at: <https://mybkgjvgnd.github.io/Fruit-306-Dataset/>.

5.2. Experimental Results

Table 3 presents a comprehensive comparison of 14 individual backbone models together with the proposed FruitEnsemble framework. The four heterogeneous backbones selected for the ensemble are highlighted in bold. DenseNet201 achieves the highest individual Top-1 accuracy, while EfficientNetB7 and ViT demonstrate superior Top-5 recall, highlighting their complementary characteristics. FruitEnsemble leverages heterogeneous ensemble aggregation to surpass all individual models.

Among individual models, DenseNet201 achieves the highest Top-1 accuracy (0.6850), indicating its strong ability to capture fine-grained texture details. EfficientNetB7 and ViT-B/16 demonstrate the best Top-5 accuracy (0.8916 and 0.8831), suggesting their effectiveness in capturing broader semantic cues. In contrast, several traditional models, such as Inceptionv3, show significantly degraded performance due to limited adaptability to high intra-class variance.

The multimodal model Qwen-VL-Plus achieves 0.5638 Top-1 accuracy but suffers from extremely high latency (1250 ms), making it unsuitable for real-time applications. In comparison, the proposed FruitEnsemble achieves 0.7049 Top-1 accuracy and 0.9150 Top-5 accuracy while maintaining an average latency of only 19.8 ms. Overall, FruitEnsemble improves the baseline accuracy by **+5.0% absolute** ($65.5\% \rightarrow 70.5\%$), demonstrating the effectiveness of the proposed hybrid architecture.

5.3. Ablation Study

To better understand the contribution of each module, we conduct ablation experiments on the key components of FruitEnsemble. Figure 5 illustrates the progressive improvement path from individual backbones to the final integrated system. Starting from the ResNet50 baseline (65.5%

Table 3. **Comprehensive Baseline Performance Analysis.** Comparison of 14 individual backbones and the proposed FruitEnsemble. The **four selected heterogeneous backbones** are highlighted in bold.[†] Average API latency per request. ^{*} Average system latency (85% direct path).

Model	Top-1 Acc	Top-5 Acc	Params (M)	FLOPs (G)	Latency (ms)
GoogleNet [31]	0.6240	0.8164	6.8	1.5	8.2
VGG16 [30]	0.5860	0.7595	138.0	15.5	22.4
Inceptionv3 [32]	0.1635	0.2763	23.8	5.7	14.1
Xception [6]	0.5830	0.7493	22.9	8.4	16.5
InceptionResnetV2 [33]	0.6734	0.8062	55.9	13.2	25.6
NasnetLarge [47]	0.2228	0.3914	88.9	23.6	38.2
ResNet50 [17]	0.6503	0.8658	25.6	4.1	12.5
DenseNet201 [21]	0.6850	0.8802	20.0	4.4	18.2
EfficientNetB7 [34]	0.6283	0.8916	66.3	37.0	45.6
ViT-B/16 [9]	0.5969	0.8831	86.6	17.6	32.4
CONVT [39]	0.5802	0.7836	28.4	6.2	15.8
Beit_large [2]	0.5373	0.6744	304.0	63.4	55.2
RVT [26]	0.5347	0.7229	102.5	24.1	36.5
Qwen-VL-Plus [36]	0.5638	-	API	API	1250 [†]
FruitEnsemble (Ours)	0.7049	0.9150	-	-	19.8[*]

Table 4. **Ablation Study on FruitEnsemble Components.**

Configuration	Heterogeneous	TTA	LLM Arb.	Top-1 Acc (%)
Baseline (ResNet50 [17])	-	-	-	65.03
+ Heterogeneous Ensemble	✓	-	-	68.50
+ Test-Time Augmentation	✓	✓	-	68.92
+ LLM Arbitration (Ours)	✓	✓	✓	70.49

All configurations use the same backbone parameters. The LLM is triggered only for samples with confidence < 0.60.

Top-1 accuracy), heterogeneous ensemble modeling significantly improves performance to 68.5%, confirming the complementarity between CNN and transformer architectures. Introducing test-time augmentation further increases accuracy to 68.9%, while the LLM-based arbitration mechanism resolves ambiguous cases and achieves the final 70.5% accuracy.

Figure 6 illustrates the effect of LLM trigger thresholds on overall accuracy. The relatively flat curve suggests that the base ensemble model has high confidence in most samples, and LLM intervention primarily benefits a small subset of difficult cases. This analysis helps determine the optimal threshold for deploying LLM arbitration in practice, trading off between accuracy gains and computational overhead.

6. Conclusion

To meet the requirements of **accurate classification and real-time efficiency** for agricultural automated sorting, this study constructed the Fruit-306 dataset containing 306 fruit

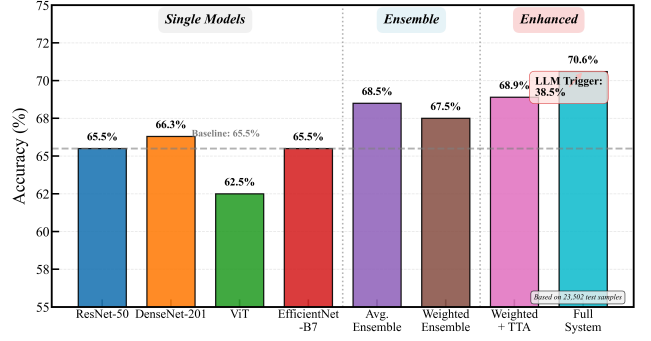


Figure 5. **Progressive performance improvement of FruitEnsemble.** Starting from individual backbones, heterogeneous ensemble aggregation, test-time augmentation, and LLM arbitration progressively improve accuracy from 65.5% to 70.5%.

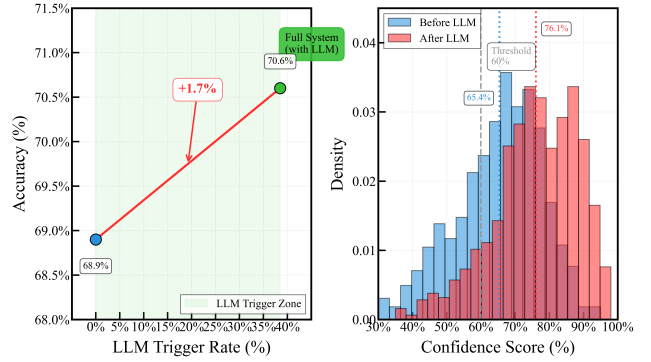


Figure 6. **Impact of LLM trigger rate on classification accuracy based on test samples.** As the LLM trigger rate increases from 0% to 40%, accuracy improves marginally from 68.90% to 70.49%, indicating that LLM arbitration provides limited gains on this dataset. The optimal trigger rate is around 38.5%, balancing computational cost and performance improvement.

categories and more than 116K images captured in real-world scenarios, and proposed the FruitEnsemble dynamic collaborative framework, which is specially designed to address the core challenges of **high cultivar similarity, large intra-class variation and imbalanced data distribution**. Through an uncertainty-aware two-stage mechanism, visual models efficiently process easy samples while the LLM arbitrates only a small number of ambiguous cases, achieving an optimal balance between accuracy and efficiency with an average latency of merely 19.8 milliseconds. Experimental results show that classification accuracy is significantly improved, with a Top-1 accuracy of 70.49% that outperforms all single models and static ensembles. The framework can accurately identify visually similar fruit cultivars and exhibits markedly enhanced robustness in real-world agricultural environments with complex lighting, occlusion, and varying ripeness levels.

References

- [1] Mounes Astani, Mohammad Hasheminejad, and Mahsa Vaghefi. A diverse ensemble classifier for tomato disease recognition. *Computers and Electronics in Agriculture*, 198: 107054, 2022.
- [2] Hangbo Bao, Li Dong, Songhao Piao, and Furu Wei. Beit: Bert pre-training of image transformers. *arXiv preprint arXiv:2106.08254*, 2021.
- [3] Peter L Bartlett and Marten H Wegkamp. Classification with a reject option using a hinge loss. *Journal of Machine Learning Research*, 9(8), 2008.
- [4] Anuja Bhargava and Atul Bansal. Classification and grading of multiple varieties of apple fruit. *Food Analytical Methods*, 14(7):1359–1368, 2021.
- [5] Md Hasan Imam Bijoy et al. Fruitvision: A benchmark dataset for fresh, rotten, and formalin-mixed fruit detection. *Data in Brief*, 61:111752, 2025.
- [6] François Chollet. Xception: Deep learning with depthwise separable convolutions. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1251–1258, 2017.
- [7] Rafael MO Cruz, Robert Sabourin, George DC Cavalcanti, and Tsang Ing Ren. Meta-des: A dynamic ensemble selection framework using meta-learning. *Pattern recognition*, 48(5):1925–1935, 2015.
- [8] Thomas G Dietterich. Ensemble methods in machine learning. In *International workshop on multiple classifier systems*, pages 1–15. Springer, 2000.
- [9] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.
- [10] Mengze Du, Fei Wang, Yu Wang, Kun Li, Wenhui Hou, Lu Liu, Yong He, and Yuwei Wang. Improving long-tailed pest classification using diffusion model-based data augmentation. *Computers and Electronics in Agriculture*, 234:110244, 2025.
- [11] Jianlong Fu, Heliang Zheng, and Tao Mei. Look closer to see better: Recurrent attention convolutional neural network for fine-grained image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4438–4446, 2017.
- [12] Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *international conference on machine learning*, pages 1050–1059. PMLR, 2016.
- [13] Disha Garg and Mansaf Alam. Smart agriculture: A literature review. *Journal of Management Analytics*, 10(2):359–415, 2023.
- [14] Mihai Gheorghe. Fruits 360: A dataset of images containing fruits and vegetables. <https://www.kaggle.com/datasets/moltean/fruits>, 2018. Accessed: 2024-05-20.
- [15] Alex Graves. Generating sequences with recurrent neural networks. *arXiv preprint arXiv:1308.0850*, 2013.
- [16] Ju He, Jie-Neng Chen, Shuai Liu, Adam Kortylewski, Cheng Yang, Yutong Bai, and Changhu Wang. Transfg: A transformer architecture for fine-grained recognition. In *Proceedings of the AAAI conference on artificial intelligence*, pages 852–860, 2022.
- [17] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016.
- [18] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.
- [19] Saihui Hou, Yushan Feng, and Zilei Wang. Vegfru: A domain-specific dataset for fine-grained visual categorization. In *Proceedings of the IEEE international conference on computer vision*, pages 541–549, 2017.
- [20] Saihui Hou, Xu Liu, and Zilei Wang. Dualnet: Learn complementary features for image recognition. In *Proceedings of the IEEE international conference on computer vision*, pages 502–510, 2017.
- [21] Gao Huang, Zhuang Liu, Laurens Van Der Maaten, and Kilian Q Weinberger. Densely connected convolutional networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4700–4708, 2017.
- [22] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. Imagenet classification with deep convolutional neural networks. *Advances in neural information processing systems*, 25, 2012.
- [23] Nan Li, Yang Yu, and Zhi-Hua Zhou. Diversity regularized ensemble pruning. In *Joint European conference on machine learning and knowledge discovery in databases*, pages 330–345. Springer, 2012.
- [24] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *Proceedings of the IEEE international conference on computer vision*, pages 2980–2988, 2017.
- [25] Xi Meng, Yingchun Yuan, Guifa Teng, and Tianzhen Liu. Deep learning for fine-grained classification of jujube fruit in the natural environment. *Journal of Food Measurement and Characterization*, 15(5):4150–4165, 2021.
- [26] Nicola Messina, Giuseppe Amato, Fabio Carrara, Claudio Gennaro, and Fabrizio Falchi. Recurrent vision transformer for solving visual reasoning problems. In *International Conference on Image Analysis and Processing*, pages 50–61. Springer, 2022.
- [27] Mehmet Alican Noyan. Uncovering bias in the plantvillage dataset. *arXiv preprint arXiv:2206.04374*, 2022.
- [28] Mihai Oltean and Horea Muresan. Fruits 360 dataset on github. 2017.
- [29] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PmlR, 2021.
- [30] Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*, 2014.

- [31] Christian Szegedy, Wei Liu, Yangqing Jia, Pierre Sermanet, Scott Reed, Dragomir Anguelov, Dumitru Erhan, Vincent Vanhoucke, and Andrew Rabinovich. Going deeper with convolutions. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1–9, 2015.
- [32] Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jon Shlens, and Zbigniew Wojna. Rethinking the inception architecture for computer vision. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2818–2826, 2016.
- [33] Christian Szegedy, Sergey Ioffe, Vincent Vanhoucke, and Alexander Alemi. Inception-v4, inception-resnet and the impact of residual connections on learning. In *Proceedings of the AAAI conference on artificial intelligence*, 2017.
- [34] Mingxing Tan and Quoc Le. Efficientnet: Rethinking model scaling for convolutional neural networks. In *International conference on machine learning*, pages 6105–6114. PMLR, 2019.
- [35] Surat Teerapittayanon, Bradley McDanel, and Hsiang-Tsung Kung. Branchynet: Fast inference via early exiting from deep neural networks. In *2016 23rd international conference on pattern recognition (ICPR)*, pages 2464–2469. IEEE, 2016.
- [36] Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, et al. Qwen2-vl: Enhancing vision-language model’s perception of the world at any resolution. *arXiv preprint arXiv:2409.12191*, 2024.
- [37] Yiqun Wang, Fahai Wang, Wenbai Chen, Bowen Lv, Mengchen Liu, Xiangyuan Kong, Chunjiang Zhao, and Zhaochen Pan. A large language model for multimodal identification of crop diseases and pests. *Scientific Reports*, 15(1):21959, 2025.
- [38] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837, 2022.
- [39] Haiping Wu, Bin Xiao, Noel Codella, Mengchen Liu, Xiyang Dai, Lu Yuan, and Lei Zhang. Cvt: Introducing convolutions to vision transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 22–31, 2021.
- [40] Miao Xu, Xuan Zhang, ChangJun Zhan, JianYu Ge, and Hua Yang. Research on citrus grading system based on machine vision. *Systems Science & Control Engineering*, 13(1): 2460443, 2025.
- [41] Zhengyuan Yang, Linjie Li, Kevin Lin, Jianfeng Wang, Chung-Ching Lin, Zicheng Liu, and Lijuan Wang. The dawn of llms: Preliminary explorations with gpt-4v (ision). *arXiv preprint arXiv:2309.17421*, 2023.
- [42] Yifan Yao, Jinhao Duan, Kaidi Xu, Yuanfang Cai, Zhibo Sun, and Yue Zhang. A survey on large language model (llm) security and privacy: The good, the bad, and the ugly. *High-Confidence Computing*, 4(2):100211, 2024.
- [43] Jason Yosinski, Jeff Clune, Yoshua Bengio, and Hod Lipson. How transferable are features in deep neural networks? *Advances in neural information processing systems*, 27, 2014.
- [44] Michael Zhang, James Lucas, Jimmy Ba, and Geoffrey E Hinton. Lookahead optimizer: k steps forward, 1 step back. *Advances in neural information processing systems*, 32, 2019.
- [45] Ning Zhang, Jeff Donahue, Ross Girshick, and Trevor Darrell. Part-based r-cnns for fine-grained category detection. In *European conference on computer vision*, pages 834–849. Springer, 2014.
- [46] Yiming Zhang, Zicheng Zhang, Xinyi Wei, Xiaohong Liu, Guangtao Zhai, and Xiongkuo Min. Illusionbench: A large-scale and comprehensive benchmark for visual illusion understanding in vision-language models. In *2025 IEEE International Conference on Multimedia and Expo (ICME)*, pages 1–6. IEEE, 2025.
- [47] Barret Zoph, Vijay Vasudevan, Jonathon Shlens, and Quoc V Le. Learning transferable architectures for scalable image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 8697–8710, 2018.